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Studierichting: Informatica

Oefeningenreeks 4A nr. 44.4

$$\begin{aligned} \int \frac{x^4 + x^3 - 3x + 5}{x^3 + 2x^2 + 2x + 1} dx &= \int \left(x - 1 \frac{-2x + 6}{x^3 + 2x^2 + 2x + 1} \right) dx \\ &= \frac{x^2}{2} - x + \int \frac{-2x + 6}{(x+1)(x^2+x+1)} dx \\ &\rightarrow \frac{-2x+6}{(x+1)(x^2+x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+x+1} \\ &\Rightarrow A = \frac{-2 \cdot -1 + 6}{1 - 1 + 1} = 8 \\ &\Rightarrow B \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) + C = \frac{7 - \sqrt{3}i}{\frac{1}{2} + \frac{\sqrt{3}}{2}i} = 2 - 4\sqrt{3}i \Rightarrow B = -8, C = -2 \\ &= \frac{x^2}{2} - x + 8 \int \frac{1}{x+1} dx + \int \frac{-4(2x+1)+2}{x^2+x+1} dx \\ &= \frac{x^2}{2} - x + 8 \ln|x+1| - 4 \ln|x^2+x+1| + 2 \int \frac{1}{x^2+x+1} dx \\ &= \frac{x^2}{2} - x + 8 \ln|x+1| - 4 \ln|x^2+x+1| + 2 \int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} d\left(x + \frac{1}{2}\right) \\ &= \frac{x^2}{2} - x + 8 \ln|x+1| - 4 \ln|x^2+x+1| + \frac{2}{\sqrt{\frac{3}{4}}} \operatorname{Bgtan} \frac{x + \frac{1}{2}}{\sqrt{\frac{3}{4}}} + c \\ &= \frac{x^2}{2} - x + 8 \ln|x+1| - 4 \ln|x^2+x+1| + \frac{4}{\sqrt{3}} \operatorname{Bgtan} \frac{2x+1}{\sqrt{3}} + c \end{aligned}$$

References